Exponential Functions and Their Graphs

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

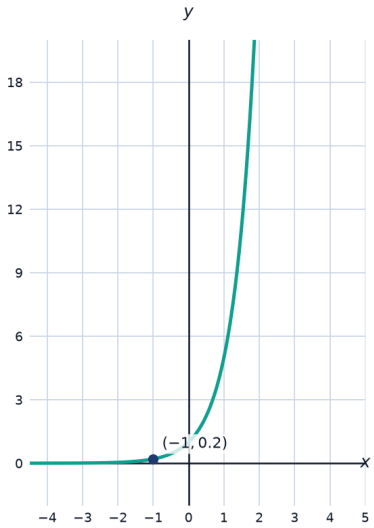
1 **[Foundational]** • **Target: Evaluating and describing**

Let $f(x) = 3 \cdot 2^x$. Evaluate:

a $f(0)$ **b** $f(3)$ **c** $f(-2)$

2 **[Foundational]** • **Target: Evaluating and describing**

For $y = 5^x$, state:



a the domain **b** the range

c the horizontal asymptote **d** the y-intercept

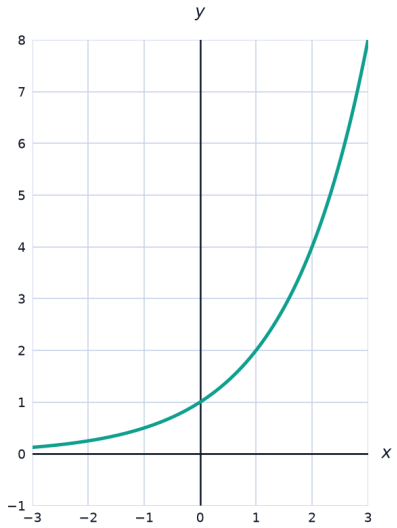
3 **[Foundational]** • **Target: Evaluating and describing**

Describe how the graph of $g(x) = 2^{x-3} + 1$ is obtained from the graph of $y = 2^x$, and state the asymptote of g.

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4 [Foundational] • Target: Identifying from graphs and applications

The graph below passes through $(0, 1)$, $(1, 2)$, and $(2, 4)$. Write the equation of the exponential function shown.

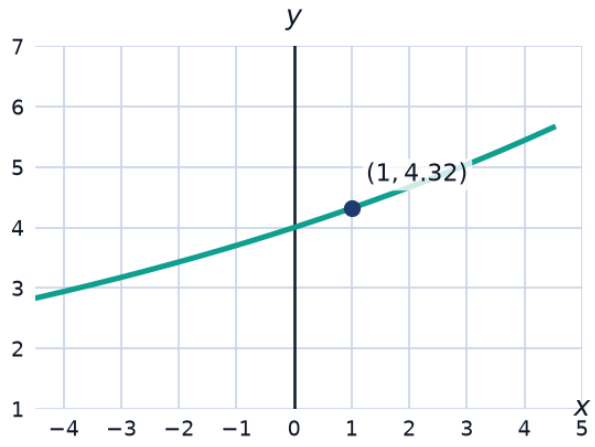


5 [Foundational] • Target: Identifying from graphs and applications

An investment of 1000 riyals grows with continuous compounding at an annual rate of 6%: $A(t) = 1000e^{0.06t}$. Find the value after 5 years, to the nearest riyal.

6 [Foundational] • Target: Identifying from graphs and applications

Determine whether each function represents growth or decay:



a $y = 4(1.08)^x$

b $y = 7(0.85)^x$

c $y = 2e^{-0.3x}$

d $y = \left(\frac{5}{4}\right)^x$

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7 [Foundational] • Target: Compound interest

2000 riyals is invested at 5% annual interest, compounded monthly, using $A = P(1 + \frac{r}{n})^{nt}$.

a Find the value after 3 years, to the nearest riyal.

b Find the value after 3 years if compounded continuously instead ($A = Pe^{rt}$), to the nearest riyal.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Compound interest

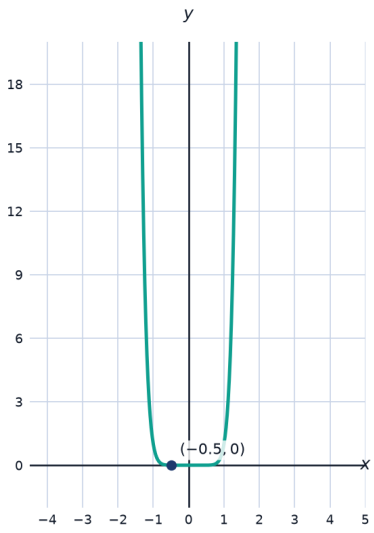
How much must be invested now at 4% annual interest compounded annually to have 10,000 riyals in 8 years? Round to the nearest riyal.

9 [Intermediate] • Target: Comparing exponential growth rates

Two investments start at 500 riyals: one grows at 4% per year, the other at 7% per year. Find the value of each after 10 years, to the nearest riyal, and their difference.

10 [Intermediate] • Target: Comparing exponential growth rates

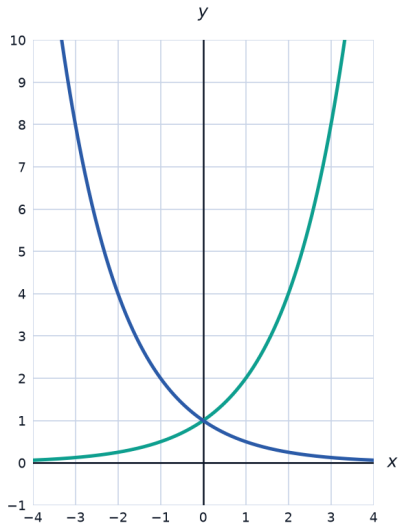
Rank the following functions from slowest to fastest growth as $x \to \infty$: $y = x^{10}$, $y = 2^x$, $y = 1.01^x$.



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11 [Intermediate] • Target: Comparing growth and decay graphically

The graphs of $y = 2^x$ (growth) and $y = \left(\frac{1}{2}\right)^x$ (decay) are shown together. Use the graph to describe the symmetry relating the two curves.



12 [Intermediate] • Target: The natural exponential base e

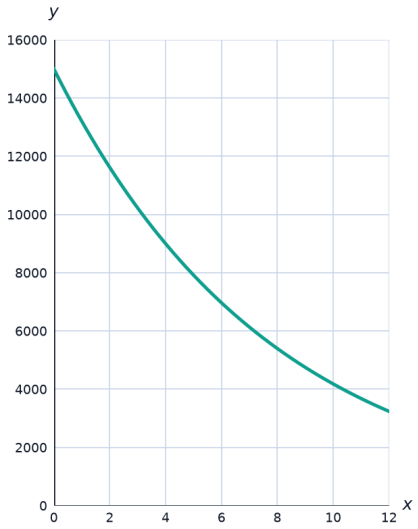
Explain what distinguishes $e \approx 2.71828$ from other exponential bases, in terms of the slope of $y = e^x$ at $x = 0$.

13 [Intermediate] • Target: The natural exponential base e

Evaluate e^0, e^1 (to 3 decimal places), and simplify $e^{\ln 5}$.

14 [Intermediate] • Target: Piecewise-exponential applications

A machine depreciates in value according to $V(t) = 15000(0.88)^t$ (riyals, years). Find its value after 5 years, and the year in which its value first drops below 5000 riyals (to the nearest year).



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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Transformations of exponential functions • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

State the horizontal asymptote and range of each transformed exponential function:

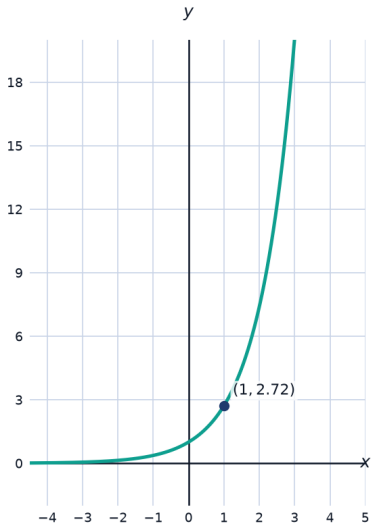
a $y = 3^x + 4$

b $y = -2^x$

c $y = 5^x - 3$

16 [Challenging] • Target: Transformations of exponential functions • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Write the equation of the function obtained by taking $y = e^x$, reflecting it across the y-axis, then shifting it down 2 units.



17 [Challenging] • Target: Comparing exponential functions numerically • Cross-Lesson: 3.01 The Unit Circle and Exact Values

For $f(x) = 3^x$, complete a short table of values at $x = -1, 0, 1, 2, 3$ and verify each value satisfies $f(x + 1) = 3f(x)$.

18 [Challenging] • Target: Comparing exponential functions numerically • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Find the inverse function $f^{-1}(x)$ and state its domain and range for $f(x) = \frac{3x - 1}{2x + 5}$.

19 [Challenging] • Target: Evaluating and describing • Cross-Lesson: 3.03 Sum and Difference Formulas

Solve the exponential equation $3^{2x - 1} = 5^{x + 2}$ for x in terms of natural logarithms.

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20 [Challenging] • Target: Identifying from graphs and applications • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Level 1 · FoundationsA sample of a radioactive isotope has an initial mass of 250 mg and decays at a continuous rate of 4.8% per day according to $M(t) = 250e^{-0.048t}$, where t is measured in days. Determine the exact daily multiplicative decay factor b in the form $M(t) = 250b^t$, and calculate the daily percentage loss rounded to two decimal places.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Evaluating and describing

If compound interest is $A = P(1 + \frac{r}{n})^{nt}$, which formula gives continuous compounding?

a. $A = Pe^{rt}$

b. $A = P/e^{rt}$

c. $A = P(1 + r)^t$

d. $A = Prt$

22 [Foundational] • Target: Identifying from graphs and applications

The graph of $y = -2f(x + 1) - 3$ applies which transformations to $y = f(x)$?

a. Left 1, stretch 2, down 3 (no reflect)

b. Right 1, reflect, stretch 2, up 3

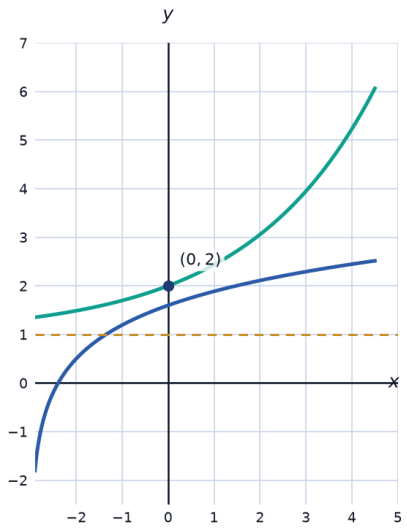
c. Right 1, stretch 2, down 3

d. Left 1, reflect over x-axis, vertical stretch 2, down 3

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23 [Foundational] • Target: Compound interest

Express $\log_3 7$ using the change of base formula with natural log.



a. $\frac{\ln 7}{\ln 3}$

b. $7\ln 3$

c. $\frac{\ln 3}{\ln 7}$

d. $\ln 7 - \ln 3$

24 [Foundational] • Target: Comparing exponential growth rates

Consider the function $f(x) = ax^2 + bx - 5$. If the graph of $f(x)$ passes through the points $(-2, 3)$ and $(1, -3)$, find the value of $a - b$.

a. 1

b. -1

c. 2

d. -2

25 [Foundational] • Target: Comparing growth and decay graphically

Let $f(x) = x^2 + 1$ and $g(x) = 3x - 2$. What is the value of $(f \circ g)(2)$?

a. 17

b. 13

c. 20

d. 16

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26 [Foundational] • Target: The natural exponential base e

A scientist is modeling the population growth of a rare species of bioluminescent fungi. The population, $P(t)$, in thousands, after t months is given by the function $P(t) = P_0 e^{kt}$, where P_0 is the initial population and k is the growth constant. Observations show that after 2 months, the population has doubled from its initial size. Furthermore, after 6 months, the population is 16,000. What is the initial population P_0?

a. 2,000

b. 4,000

c. 8,000

d. 1,000

27 [Foundational] • Target: Piecewise-exponential applications

Determine all continuous even functions f satisfying $f(f(x)) = x$.

a. $f(x) = x^2$

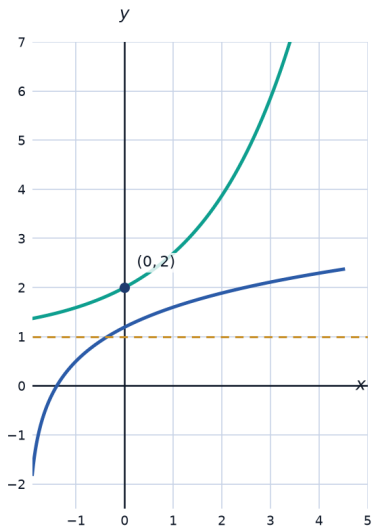
b. $f(x) = 1/x$

c. $f(x) = |x|$

d. $f(x) = x$ or $f(x) = -x$

28 [Foundational] • Target: Transformations of exponential functions

Find the number of real-valued functions $f : \{1, 2, 3\} \to \{1, 2, 3\}$ satisfying $f(f(x)) = f(x)$ for all x.



a. 9

b. 10

c. 27

d. 6

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: Comparing exponential functions numerically**

Consider the function $f(x) = 2x + 5$. Which of the following represents the inverse function $f^{-1}(x)$?

a. $f^{-1}(x) = \frac{x-5}{2}$

b. $f^{-1}(x) = \frac{x+5}{2}$

c. $f^{-1}(x) = \frac{2}{x} - 5$

d. $f^{-1}(x) = 2x - 5$

30 **[Intermediate]** • **Target: Evaluating and describing**

For $f : \mathbb{Z} \to \mathbb{Z}$, if $f(m+n) + f(mn-1) = f(m)f(n) + 1$ for all integers, find $f(2)$.

a. 1

b. 3

c. 4

d. 2

31 **[Intermediate]** • **Target: Identifying from graphs and applications**

Consider the function $g(x) = x^2 - 6x + 5$. What is the x-coordinate of the vertex of the parabola represented by this function?

a. 3

b. -3

c. 6

d. -6

32 **[Intermediate]** • **Target: Compound interest**

Solve: $f(x) = f(-x)$ for all x where $f(x) = x^3 + ax$. Find a.

a. $a = -1$

b. Any a

c. $a = 0$

d. $a = 1$

33 **[Intermediate]** • **Target: Comparing exponential growth rates**

Classify all solutions of $f(f(x)) = f(x)$ (idempotents) for $f(x) = x^2$.

a. $x = 1$ only

b. $x = 0$ or $x = 1$

c. All $x \geq 0$

d. $x = 0$ only

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34 [Intermediate] • Target: Comparing growth and decay graphically

Given $f(x) = \frac{1}{2-x}$, find $f^{(n)}(x)$ (the n-th iterate).

a. $f^n(x) = nx + 1$

b. $f^n(x) = f(x)$ for all n

c. $f^n(x) \to \infty$

d. Cycles with period 3; $f^3(x) = x$

35 [Intermediate] • Target: The natural exponential base e

Find all k such that $f(x) = kx^2 + (k+2)x + 2k - 1$ is always negative.

a. $k < -1$

b. All $k < 0$

c. $k < -\frac{1}{3}$ and discriminant < 0

d. No such k exists for all x

36 [Intermediate] • Target: Piecewise-exponential applications

What is the half-life formula for exponential decay?

a. $A(t) = A_0(2)^{t/h}$

b. $A(t) = A_0 e^{-t}$

c. $A(t) = A_0/2^t$

d. $A(t) = A_0\left(\frac{1}{2}\right)^{t/h}$

37 [Intermediate] • Target: Transformations of exponential functions

Find all real solutions to $f(2^x) = f(3^x)$ where $f(t) = t^2 - t$.

a. $x = 0$

b. $x = 1$

c. No solution

d. $x = \log_6 6$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Comparing exponential functions numerically • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Which is an exponential decay?

a. $y = 3^x$

b. $y = 0.4^x$

c. $y = e^x$

d. $y = 10^x$

39 [Challenging] • Target: Evaluating and describing • Cross-Lesson: 3.03 Sum and Difference Formulas

Write using natural log: $\log_b x =$

a. $\frac{\ln b}{\ln x}$

b. $\ln x - \ln b$

c. $\ln x \cdot \ln b$

d. $\frac{\ln x}{\ln b}$

40 [Challenging] • Target: Identifying from graphs and applications • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

If $f(x) = \arcsin(\sin x)$, what is the period of f?

a. 2π

b. π

c. $\pi/2$

d. Not periodic

41 [Challenging] • Target: Compound interest • Cross-Lesson: 4.01 Circles and Parabolas

A function satisfies $f(1) = 2, f(2) = 5, f(3) = 10$. Identify the pattern.

a. $f(n) = n + 1$

b. $f(n) = 2^n$

c. $f(n) = n^2 + 1$

d. $f(n) = 3n - 1$

42 [Challenging] • Target: Comparing exponential growth rates • Cross-Lesson: 4.02 Ellipses

Find the range of $f(x) = \frac{1}{x^2 + 1}$ for all real x.

a. $(0, 1)$

b. $(0, 1]$

c. $[0, 1]$

d. $[0, 1)$

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43 [Challenging] • Target: Comparing growth and decay graphically • Cross-Lesson: 4.03 Hyperbolas

Given $f(x) = x^n$ for integer $n \geq 1$, find the number of real solutions to $f(f(x)) = x$.

a. Always n

b. Depends on n: n^2 solutions in general

c. Always 1

d. Always 2

44 [Challenging] • Target: The natural exponential base e • Cross-Lesson: 5.01 Vectors: Components and Operations

Find all x satisfying $f(x) > 0$ where $f(x) = (x-1)(x-2)(x-3)$.

a. 13

b. $x < 1$ or $2 < x < 3$

c. $x > 1$

d. $x < 3$ only

45 [Challenging] • Target: Piecewise-exponential applications • Cross-Lesson: 5.02 The Dot Product

If f is monotonically increasing, prove f^{-1} is also increasing.

a. True only if f is linear

b. True only if f is differentiable

c. If $a < b$ then $f^{-1}(a) < f^{-1}(b)$ — otherwise $f(f^{-1}(a)) \geq f(f^{-1}(b))$ contradicting monotonicity

d. Not necessarily true

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